

Arnav Gupta

Independent Researcher

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Research Program & Overview

Independent researcher emerging in pure mathematics—optimization, matrix analysis, and calculus—with published work in machine learning theory and existentialist philosophy. Three ML/NLP preprints establish the empirical and theoretical foundation from which this direction emerged: a spectral saturation index for few-shot stopping rules (arXiv), a controlled documentation of verifier exploitation in NLI-guided refinement (Research Square), and a Wedin-type alignment bound for norm-constrained low-rank updates (Zenodo). A fourth preprint (Zenodo) critically re-examines the argumentative structure of Camus’s *Myth of Sisyphus*. All work conducted independently from Nepal.

Preprints & Publications

1. Singular-Subspace Alignment in Norm-Constrained Asymmetric Low-Rank Updates.

Arnav Gupta. *Zenodo preprint*, 2026. [doi:10.5281/zenodo.22845046]

- Proved a **Wedin-type alignment bound** for asymmetric low-rank updates $X = BA$ under a Frobenius-norm constraint: alignment error $\leq \beta c / \delta_r(G, \tilde{G})$, where β is the gradient-Lipschitz constant and c the update budget.
- Showed that factorized second-order stationarity selects the leading r singular subspaces of the gradient exactly under the affine linearization of a smooth matrix loss.
- Established that for $r = 1$, exact alignment can fail outside the perturbative regime at a genuine factorized second-order stationary point.

2. The Geometry of Saturation: Effective Rank Predicts When Labels Stop Helping in Few-Shot Classification.

Arnav Gupta. *arXiv preprint*, 2026 (v2). [doi:10.48550/arXiv.2606.24903]

- Introduced the **spectral saturation index** $S(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$ —a closed-form, label-free scalar grounded in high-dimensional covariance estimation that predicts diminishing returns in few-shot label acquisition in $\mathcal{O}(d^3)$ time.
- Validated across **49 real tasks on 3 frozen backbones** (PCA-50, CLIP ViT-B/32, DINOv2 ViT-S/14) across 6 datasets (**492 doubling-pair observations**).
- Fixed threshold $\tau = 0.02$ classifies stop/continue decisions with **AUC = 0.787**, achieving **100% recall on gains** ($\Delta A > 1\%$).

3. Verifier Exploitation in NLI-Guided Iterative Refinement: A Controlled Empirical Analysis.

Arnav Gupta. *Research Square preprint*, 2026. [doi:10.21203/rs.3.rs-10187492/v1]

- First controlled empirical documentation of verifier exploitation in zero-gradient, prompt-only LLM refinement: revision cycle inflated SummaCConv by +0.185 while collapsing BARTScore_{s→d} by −2.566 nats ($W = 0$, $p = 2.68 \times 10^{-83}$) across all 498 test instances.
- Identified truncation-exploiting content removal; developed an annotation-free 3-condition detection protocol for NLI verifier pipelines.
- Proposed AnchorSum, achieving a 6.3% relative SummaCConv inconsistency reduction over Multi-News ($p = 4.49 \times 10^{-28}$) and outperforming fine-tuned models across all LLM-as-judge dimensions.

4. Assertion and Argument: Reexamining Happiness in Camus’s *The Myth of Sisyphus*.

Arnav Gupta. *Zenodo preprint*, 2026. [doi:10.5281/zenodo.22843907]

- Argues that Camus’s own conclusion enacts the very “philosophical suicide” he critiques in Kierkegaard, Chestov, and Husserl—the transition from Sisyphus’s acceptance of fate to his proclaimed happiness exceeds what the essay’s argument demonstrates.
- Examines and rejects the two strongest defenses of the conclusion, contending that neither bridges the identified inferential gap.

Education

Sagarmatha Secondary School, Mirchaiya, Siraha, Nepal
Grade 10 GPA: 3.82 / 4.00 (*Outstanding*)

Expected SEE: 2027

Fellowships & Research Experience

Suvidha Foundation
Machine Learning Intern (Remote)

Sept 2025 – Nov 2025

- Researched multi-document abstractive summarization with transformer-based models under mentorship of a PhD in NLP.
- Built proof-of-concept summarization pipeline with hierarchical context aggregation across multiple source documents.

Mathematics Coursework

MIT OpenCourseWare — *Self-Directed*

- **18.01 — Single Variable Calculus** (*Completed*): Differential and integral calculus with rigorous problem-solving.
- **18.06 — Linear Algebra** (*Completed*): Matrix theory, eigendecomposition, singular value decomposition (SVD), orthogonality, and applications to machine learning.
- **18.02 — Multivariable Calculus** (*In Progress*): Vector calculus, partial derivatives, multiple integrals, line & surface integrals, and gradient fields.